

28. Group Characters and Ideal Theory.

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3. E. NOETHER, Göttingen: Group characters and ideal theory.

The Frobenius theory of group characters – that is, of representations of finite groups – is conceived as the ideal theory of a completely reducible ring, the group ring. Therefore the isomorphism theorems for the additive decomposition of a completely reducible ring are developed in general; the representation as a direct sum of directly indecomposable, simple ideals is unique up to isomorphism, and the indecomposable one-sided ideals belonging to the same two-sided ideal are mutually isomorphic. If therefore isomorphic ideals are combined into one class, then there are as many distinct indecomposable ideal classes as there are two-sided indecomposable ideals.

In the specialization to completely reducible systems of hypercomplex quantities – in particular to the group ring – indecomposable ideal classes and irreducible representation classes correspond mutually and uniquely. Furthermore, two-sided indecomposable ideals and characters correspond; and the number of additively indecomposable components of a two-sided ideal is equal to the rank of the indecomposable ideals. Thus the Frobenius theory has been classified within this framework.

A detailed presentation is to appear in the Math. Ann.